

Hyunwoo Shin

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EDUCATION

Virginia Polytechnic Institute and State University

Ph.D. in Industrial & Systems Engineering

Blacksburg, VA

Aug 2021 – May 2026

Korea Aerospace University

M.S. in Logistics, School of Air Transport, Transportation, and Logistics

Goyang, South Korea

Mar 2018 – Feb 2020

Korea Aerospace University

B.S. in Logistics, School of Air Transport, Transportation, and Logistics

Goyang, South Korea

Mar 2012 – Feb 2018

PUBLICATIONS

- [1] **H. Shin**, S. Tunc, X. Chen, J. M. Reynolds. “A Comprehensive Simulation Framework for Evaluating U.S. Lung Allocation Policies.” (*Major Revision at Health Care Management Science*)
- [2] **H. Shin**, S. Tunc, P. Afeche, M. Begen, B. Sandikci, F. Murillo, B. Hansen, M. Claasen, G. Sapisochin. “An In-depth Analysis of Organ Offer Decisions in the Canadian Liver Transplant System.” (*Working paper*)
- [3] **H. Shin**, S. Tunc, X. Chen, J. M. Reynolds. “Forecasting Dynamic Allocation Priority in Organ Transplantation.” (*Working paper*)
- [4] Y. Ha, **H. Shin**, J. Mueller. “Scalable Traffic Signal Control for Urban Mobility: A Subspace-based Optimization Approach Using Deep Reinforcement Learning.” (*Working paper*)
- [5] **H. Shin**, and J. Chae. “A Performance Review of Collision-Free Path Planning Algorithms.” *Electronics*, 9(2), 316, 2020. DOI: 10.3390/electronics9020316
- [6] H.-Y. Lee, **H. Shin**, and J. Chae. “Path Planning for Mobile Agents Using a Genetic Algorithm with a Direction Guided Factor.” *Electronics*, 7(10), 212, 2018. DOI: 10.3390/electronics7100212
- [7] M. Kim, **H. Shin**, and J. Chae. “Merge Control using Reserve Ahead Point in Baggage Handling Systems.” *Journal of the Society of Korea Industrial and Systems Engineering*, 40(2), 60–67, 2017.

RESEARCH EXPERIENCE

MD Anderson Cancer Center

Incoming Postdoctoral Researcher

Houston, TX

Starting July 2026

National Renewable Energy Laboratory

Summer Intern (Mentor: Yunsoo Ha; Supervisor: Julianne Mueller)

Remote

June 2025 – Aug 2025

- Simulation optimization for traffic simulation.
- Finding the subspace for optimizing via the simulation model using a deep reinforcement learning model.

Virginia Tech

Graduate Research Assistant (Advisor: Sait Tunc)

Blacksburg, VA

Spring 2022, Fall 2024, Fall 2025

- Optimized U.S. lung allocation policy using high-performance computing and discrete-event simulation.
- Verified and validated a liver and kidney transplant simulation model via Bayesian optimization approaches and ASTRO-DF algorithm.
- Developed a simulation model for U.S. lung transplant allocation and processed large-scale clinical data.
- Investigated modeling techniques to estimate lung allocation scores under uncertainty.

Korea Aerospace University

Researcher & Research Assistant (Advisor: Junjae Chae)

Goyang, South Korea

Mar 2016 – Jul 2021

- Collaborated on a multi-shuttle control logic for an automated container yard (CyberLogitec & Korea Maritime Institute).
- Developed job assignment algorithms for a new business model (Nexen Tire & SL Solution).

- Analyzed automation logic for gantry cranes using simulation (Hanjin Logistics Institute).
- Evaluated methodologies for UAV path planning; conducted efficiency and performance analyses.
- Developed a vehicle routing problem (VRP) solution incorporating real map distances and operational constraints (SK Telecom & SL Solution).
- Built an unmanned logistics system with UAV pathfinding via Genetic Algorithms (Ministry of Land, Infrastructure and Transport).
- Analyzed a baggage handling system merging control logic using AutoMod.

TEACHING EXPERIENCE

Virginia Tech Blacksburg, VA
Instructor of Record, Department of Industrial & Systems Engineering Spring 2026
 – **Undergraduate Course:** Logistics Engineering

Virginia Tech Blacksburg, VA
Teaching Assistant, Department of Industrial & Systems Engineering and Department of Statistics 2022 – 2025
 – **Undergraduate Courses (ISE):** Deterministic Operations Research, Logistics Engineering, Data Management, Statistical Quality Control
 – **Undergraduate Course (Statistics):** Statistics for Engineers
 – **Graduate Courses (ISE):** Random Process, Facilities Planning

Korea Aerospace University Goyang, South Korea
Instructor of Record Spring 2021
 – **Undergraduate Course:** Operations Research I

CONFERENCE PRESENTATIONS

- **INFORMS Annual Meeting 2025.** H. Shin, X. Chen, and S. Tunc. “Optimizing U.S. Lung Allocation Policy via a Multi-Objective Trust-Region-based Simulation Optimization Algorithm.”
- **INFORMS Annual Meeting 2024.** H. Shin, X. Chen, and S. Tunc. “Calibration of Simulation Models for Organ Allocation Using Conformal Prediction Concepts.”
- **INFORMS Annual Meeting 2023.** H. Shin, X. Chen, and S. Tunc. “Forecasting Organ Transplant Allocation Scores Using Machine Learning Models.”

HONORS & AWARDS

- **Paul E. Torgersen Graduate Student Research Excellence Award, Top Ten Finalist, Virginia Tech** 2026
- **ISE Graduate Student Travel Awards, Virginia Tech** 2023–2025
- **GPSS Travel Fund Program, Virginia Tech** 2025

TECHNICAL SKILLS

Languages: Python, C++, R, Java

Software Tools: AutoMod, Arena, ExtendSim, CPLEX, Gurobi

Methodologies: Simulation Optimization, Bayesian Optimization, Surrogate Modeling, Machine Learning, Deep Reinforcement Learning, Bayesian Inference

Operating Systems: Linux (Ubuntu/HPC), Windows (WSL)

SELECTED COURSEWORK

- **Probability & Statistics:** ISE 5034 (*Math Probability & Statistics for ISEs*), ISE 5984 (*Stat Learning and Data Sci*), CS 5525 (*Data Analytics*), STAT 5114 (*Statistical Inference*), STAT 5444 (*Bayesian Statistics*), STAT 6105 (*Measure & Probability*), STAT 6474 (*Advanced Topics in Bayesian Statistics*), STAT 6544 (*Surrogate Modeling*)
- **Simulation & Stochastic Modeling:** ISE 5424 (*Simulation I*), ISE 6494 (*Advanced Simulation*), ISE 5414 (*Random Process*), ISE 6464 (*Queueing Networks*)
- **Optimization:** ISE 5405, 5406 (*Optimization I & II*)

ACADEMIC SERVICE

- **Session Chair:** 2024, 2025 INFORMS Annual Meeting
- **VT INFORMS Student Chapter:** VP Event (2025–2026)